Model Interpretability: Understanding Neural Network Decisions - 2

Tuesday, 27 October 2026 12:36 AM

Agenda

- LIME revision
- SHAP
  - Shapley Values
  - Code Implementation
- LIME vs. SHAP

LIME → Local (Explain about 1 data point.) Interpretation (↓ to understand) Model Agnostic (↓ Works with any model.) Explanations

Advantages:

1) It is model agnostic.
2) It can be used for tabular, image, text
3) Use a simple surrogate model.

α → ① x → ② (x+y)

Disadvantages:

1) Sensitive to neighborhood
2) Sensitive to hyperparameters α, ε, K.
   ↑ radius ↑ noise ↑ # of top features.
3) Uses distance as a metric to find locality of interest.
   ↳ different distances can lead to different locality of interest.

Q. In production we want to use an algorithm to explain the predictions of the NN. There is a threshold of 50 ms.

50 ms?

1 NN → main model.

1 DP → ① find locality of interest → $\vec{x_q}$

② Generate predictions for each synthetic point using the main model.

multiple calls.

→ surrogate model

model ... 4 ③ Build a ...

↓

Simple LR regression model.

Log Reg.

Lin Reg.

Decision Tr XGBoost.

NN → ? (Black box model).

Quiz:-

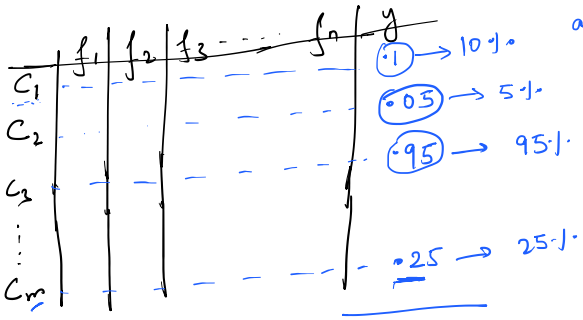
A team wants to explain individual predictions of a fraud-detection model that must return an explanation within 50 milliseconds for every transaction, in production, at high volume. Is LIME a good fit?

* SHAP

Shapley Additive Explanations.

Model → Predict default probability of customers.

Classification.



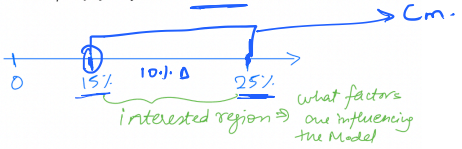
avg → sum($P_1, P_2, P_3 \dots P_m$)/m.

Baseline Score. (avg score) for a customer defaulting is 15%.

Cm → 25%.

Suppose for a person B, the model predicts 25%. then why model prediction for person B differs from our baseline ? (15.1.)

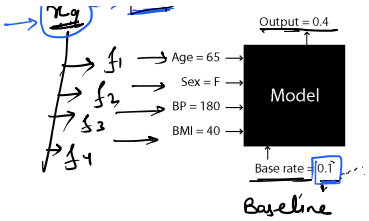
↳ this is what SHAP measures



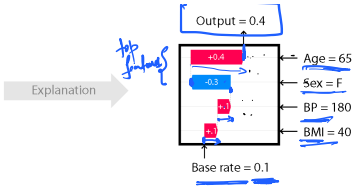
SHAP

→ 4.

... trend higher.



Baseline



red → pushes your ... up

blue → pushes your pred lower.

Δ 0.4 - 0.1 = 0.3.

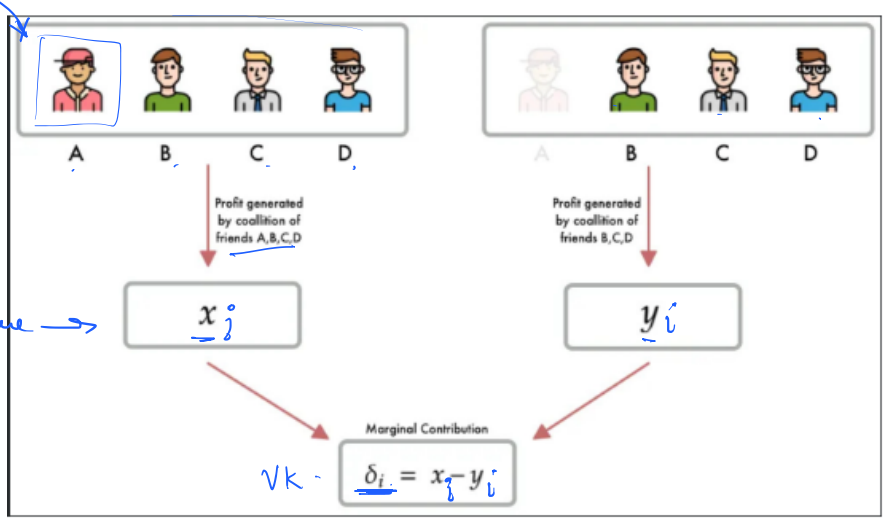
$(0.1 + 0.1 + 0.1 - 0.3 + 0.4) \rightarrow 0.4$

Game of Cricket

VK.

1 Match.

n Matches.



n Marginal Contributions

$$\text{Avg MC} = \frac{1}{n}(MC_1 + MC_2 + MC_3 - - - - + MC_n).$$

$\{A, B, C\} \rightarrow 55$

$\{B, C\} \rightarrow 40$

MC for A → 55 - 40

= 15.

Shapely Values.

tennis

10 points 12 points.

Combined is 23.

$f_1 + f_2$

EXAMPLE 1. Suppose that there are two players and $v(\{1\}) = 10$, $v(\{2\}) = 12$ and $v(\{1,2\}) = 23$. There are two possible orders of arrival: (1) first 1 then 2, and (2) first 2 then 1.

Scenario 1 If 1 comes first and then 2, 1's contribution is $v(\{1\}) = 10$; when 2 arrives the surplus increases from 10 to $v(\{1,2\}) = 23$ and therefore 2's marginal contribution is $v(\{1,2\}) - v(\{1\}) = 23 - 10 = 13$.

Scenario 2. If 2 comes first and then 1, 2's contribution is $v(\{2\}) = 12$; when 1 arrives the surplus increases from 12 to $v(\{1,2\}) = 23$ and therefore 1's marginal contribution is $v(\{1,2\}) - v(\{2\}) = 23 - 12 = 11$.

Thus we have the following table:

Probability	Order of arrival	1's marginal contribution	2's marginal contribution
$\frac{1}{2}$	first 1 then 2	10	13
$\frac{1}{2}$	first 2 then 1	11	12

Thus 1's expected marginal contribution is: $\frac{1}{2}10 + \frac{1}{2}11 = 10.5$ and 2's expected marginal contribution is $\frac{1}{2}13 + \frac{1}{2}12 = 12.5$. This is the Shapley value: $x_1 = 10.5$ and $x_2 = 12.5$.

$v_1 \to 10$

$v_2 \to 13$

$v_2 \to 12$

$v_1 \to 11$

Player 1 MC $= \frac{1}{2} \times 10 + \frac{1}{2} \times 11 = 10.5$

Player 2 MC $= \frac{1}{2} \times 13 + \frac{1}{2} \times 12 = 12.5$

$f_1\ f_2\ f_3$

ST → Schwag

$v(\{1\}) = 100$, $v(\{2\}) = 125$, $v(\{3\}) = 50$, $v(\{1,2\}) = 270$, $v(\{1,3\}) = 375$, $v(\{2,3\}) = 350$ and $v(\{1,2,3\}) = 500$

Probability	Order of arrival	1's marginal contribution	2's marginal contribution	3's marginal contribution
$\frac{1}{6}$	first 1 then 2 then 3: 123	$v(\{1\}) = 100$	$v(\{1,2\}) - v(\{1\}) = 270 - 100 = 170$	$v(\{1,2,3\}) - v(\{1,2\}) = 500 - 270 = 230$
$\frac{1}{6}$	first 1 then 3 then 2: 132	$v(\{1\}) = 100$	$v(\{1,2,3\}) - v(\{1,3\}) = 500 - 375 = 125$	$v(\{1,3\}) - v(\{1\}) = 375 - 100 = 275$
$\frac{1}{6}$	first 2 then 1 then 3: 213	$v(\{1,2\}) - v(\{2\}) = 270 - 125 = 145$	$v(\{2\}) = 125$	$v(\{1,2,3\}) - v(\{1,2\}) = 500 - 270 = 230$
$\frac{1}{6}$	first 2 then 3 then 1: 231	$v(\{1,2,3\}) - v(\{2,3\}) = 500 - 350 = 150$	$v(\{2\}) = 125$	$v(\{2,3\}) - v(\{2\}) = 350 - 125 = 225$
$\frac{1}{6}$	first 3 then 1 then 2: 312	$v(\{1,3\}) - v(\{3\}) = 375 - 50 = 325$	$v(\{1,2,3\}) - v(\{1,3\}) = 500 - 375 = 125$	$v(\{3\}) = 50$
$\frac{1}{6}$	first 3 then 2 then 1: 321	$v(\{1,2,3\}) - v(\{2,3\}) = 500 - 350 = 150$	$v(\{2,3\}) - v(\{3\}) = 350 - 50 = 300$	$v(\{3\}) = 50$

f_1 Thus 1's expected marginal contribution is: $\frac{1}{6}(100 + 100 + 145 + 150 + 325 + 150) = \frac{970}{6}$ ✓

f_2 2's expected marginal contribution is $\frac{1}{6}170 + \frac{1}{6}125 + \frac{1}{6}125 + \frac{1}{6}125 + \frac{1}{6}125 + \frac{1}{6}300 = \frac{970}{6}$ ✓

f_3 3's expected marginal contribution is $\frac{1}{6}230 + \frac{1}{6}275 + \frac{1}{6}230 + \frac{1}{6}225 + \frac{1}{6}50 + \frac{1}{6}50 = \frac{1060}{6}$ ✓

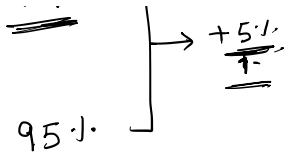
The sum, of course, is $\frac{3000}{6} = 500 = v(\{1,2,3\})$

Prediction with feature $f_1 + f_2$. → P_1

Prediction with feature f_2 → P_2

$P_1 - P_2$. → MC f_1

90%



Quiz 2.

70% Δ66%

SHAP values for one prediction are guaranteed to add up exactly to (the model's output for that instance) minus (the average model output over a baseline). Why does this matter in practice?

10%

Ans - The attributions become a complete, additive breakdown of the prediction

Q. We need to choose an algo for explainability for a healthcare model under regulatory review. Which model will we choose?

A. Due to additivity guarantee, SHAP is preferred.